

CVE-2026-32740: RCE in a PIE Next.js sharp/libheif Stack

CVE-2026-32740: RCE in a PIE Next.js sharp/libheif Stack - Adrian Tiron, Fortbridge.

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Content

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Motivation

[Hacktron's Hacking OpenAI research](#) was one of the projects that pushed us to investigate libheif exploitation more deeply. It showed what can happen when a memory-safety bug in an image decoder sits behind an ordinary web upload. Their article described the wider attack chain, but the low-level details needed to reproduce the exploit were not published. It is related prior art rather than the same vulnerability or application stack: our work targets CVE-2026-32740 in a deliberately vulnerable Next.js and sharp lab.

We have always liked the idea of reaching native memory corruption through a web application, so we decided to start from the crash and work through the exploitation ourselves. The project brought together areas that are often treated separately: web application security, memory-corruption fuzzing, GDB debugging, allocator behaviour, and binary exploitation. The OWASP Top 10 remains a useful baseline, but modern application security cannot stop there when a web route can pass attacker-controlled files through a large native-code dependency chain.

We first built the exploit against the Ubuntu setup. We then repeated the work against Debian to test our assumptions and make the proof of concept more robust across different native stacks.

CVE-2026-32740 executive summary

CVE-2026-32740 is a heap-buffer overflow in `HeifPixelImage::copy_image_to()`, the libheif function that places decoded HEIF or AVIF tiles into a grid image. libheif stores the image as separate planes. The Y plane contains brightness samples at full resolution. The Cb and Cr planes contain blue and red colour-difference samples. In 4:2:0 format, Cb and Cr have half as many columns and half as many rows as Y, with odd values rounded up.

The vulnerable rounding

The vulnerable code calculates the chroma destination row and copy height separately. Both calculations round up, but the copy-height check does not account for the already rounded destination row:

```
copy_height = min(src_height,
                  channel_height(h - y0, chroma, channel));
ys = channel_height(y0, chroma, channel);

for (py = 0; py < copy_height; py++)
    memcpy(out_data + (ys + py) * out_stride, ...);
```

Our payload stacks four tiles, each 33 pixels high, in a 132-pixel canvas. The complete Cb and Cr planes have $\text{ceil}(132 / 2) = 66$ rows, numbered 0 to 65. The fourth tile starts at Y row 99, so libheif calculates $ys = \text{ceil}(99 / 2) = 50$. It independently calculates $\text{copy_height} = \text{ceil}(33 / 2) = 17$. The loop writes chroma rows 50 through 66, and row 66 is outside the allocation.

The canvas is 116 pixels wide. An 8-bit 4:2:0 chroma row is therefore $\text{ceil}(116 / 2) = 58$ bytes. As a result, the malformed fourth tile supplies every byte in that final row, giving the attacker one controlled 58-byte write across the end of the Cb or Cr allocation.

From one extra row to code execution

A crash alone did not make the issue remotely exploitable. The final chain needed three additional capabilities:

- A separate returned-pixel disclosure to recover the randomized libvips base remotely.
- A constrained `write-what-where` primitive, so the exploit can select both the 58 bytes and their destination.
- A control-flow target that executes before another worker can disturb the corrupted state.

The write-what-where is constrained because the row length, object layout, and addresses are specific to the tested build. It is still a write-what-where: the Cr pixels provide the bytes, and a forged entry in libheif's plane map provides the destination. The first chosen-address copy overwrites `memcpy@G0T`. The next row enters a native-library loader inside the already identified libvips image and loads a shared object submitted through the image-upload route.

Tested application and dependency stack

The proof targets a deliberately vulnerable Next.js lab with two routes. `POST /api/upload` saves an uploaded file in the application's `uploads/` directory. `GET /api/optimize?file=NAME` opens a saved file with sharp, converts it to PNG, and returns that PNG to the client. AVIF decoding happens inside sharp's bundled libvips and libheif libraries.

- Next.js 15.5.23
- sharp 0.34.4
- bundled libvips 8.17.2
- bundled libheif 1.20.2
- Ubuntu profile: Node.js 25.8.1, PIE `ET_DYN`, glibc 2.43
- Debian 13 profile: Debian APT Node.js 20.19.2, PIE `ET_DYN`, glibc 2.41
- Linux x86-64 with ASLR and NX enabled

The Node executable is position independent, while sharp and its bundled native libraries are the normal prebuilt packages for this tested platform. The exploit does not need the randomized Node base because its control-flow target is inside libvips.

The application route itself is intentionally permissive. In particular, it preserves an uploaded ELF shared object under the image-looking name `x.jpg`. A real deployment with strong magic validation, generated storage names, or isolated object storage may prevent that staging step even if its decoder remains vulnerable.

Why CVE-2026-32740 needed more than a crash

The grid bug gives us one controlled 58-byte write immediately after a chroma buffer. On its own, that normally causes a crash. Four developments were essential to turn it into repeatable remote code execution:

- **Find the current process:** a separate AVIF makes the optimisation route return libvips pointers in PNG pixels, which reveals the randomized libvips base.
- **Choose the destination:** the 58-byte overflow preserves the adjacent glibc chunk header and redirects libheif's Cr-plane lookup to a forged map entry. This creates the constrained write-what-where.
- **Trigger immediately:** the chosen write replaces `memcpy@GOT` during the image-copy loop. The very next row uses the changed GOT entry, leaving almost no delay between corruption and control-flow hijack.
- **Reuse a compatible native loader:** the bundled libvips image contains an internal copy of GLib's `g_module_open_full`. This GModule routine wraps `dlopen`, loading a native shared library from a file path. The hijacked copy already puts the controlled path `uploads/x.jpg` in its first argument register. When the loader opens that file, the shared object's constructor runs, without requiring a ROP chain or a Node code address.

`g_module_open_full` is one of the essential tricks. A random code address would not be enough. We needed code inside a module whose randomized base was already disclosed,

and it had to tolerate the register state at the hijacked call site. The routine is at offset `0x3e995e` in the exact bundled libvips artifact. The exploit adds that profile-verified offset to the recovered libvips base, so every runtime code address used by the final control-flow step comes from the same disclosed module.

The final CVE-2026-32740 RCE chain

1. Upload a malicious library

On the attacker machine, the exploit compiles a small shared object. Its constructor runs the fixed proof command `/usr/bin/id` and returns the command's output to the listener. The target does not compile anything.

The application accepts the ELF through its ordinary image-upload route with the filename `x.jpg` and media type `image/jpeg`, then stores it as `uploads/x.jpg`. The short name is important because the later payload has only 16 bytes for the complete null-terminated library path.

2. Bypass ASLR using returned pixels

Address Space Layout Randomisation (ASLR) gives libvips a new base address whenever the target starts. The exploit uses the two lab routes in sequence:

- `POST /api/upload` stores a specially prepared ASLR-leak AVIF under a random filename.
- `GET /api/optimize?file=NAME` asks sharp to decode that saved AVIF and convert it to PNG.
- The optimisation route returns the PNG bytes in the HTTP response. In this image layout, some alpha-channel bytes come from uninitialised native memory and include pointers into the loaded libvips library.

This disclosure AVIF is separate from the final grid-overflow AVIF. The exploit repeats the upload and optimisation requests until one returned PNG supports exactly one profile and one libvips base. Later, it uploads the final AVIF through the same upload route and calls the same optimisation route again to trigger the overflow.

A real response contained these three shared values. Each pointer refers to a known slot in the bundled libvips build, so subtracting its file offset recovers the same page-aligned base:

```
0x7f6be25c3120 - 0x1c3120 = 0x7f6be2400000
0x7f6be25be1b0 - 0x1be1b0 = 0x7f6be2400000
0x7f6be25c3130 - 0x1c3130 = 0x7f6be2400000
```

The Ubuntu and Debian profiles use the same bundled libvips binary. In both environments, the returned pixels expose pointers at `libvips_base + 0x1be1b0`, `libvips_base + 0x1c3120`, and `libvips_base + 0x1c3130`. Subtracting any corresponding offset reveals the candidate randomized library base. Because these three anchors are shared, they confirm the libvips

binary and recover its base, but cannot distinguish the Ubuntu native stack from the Debian native stack.

To distinguish the two tested profiles, the exploit requires one more leaked pointer. Ubuntu responses reliably contain a pointer at `libvips_base + 0x33cd0c`, while Debian responses contain one at `libvips_base + 0x1de9c0`. These are offsets from the current libvips base, not fixed runtime addresses. ASLR changes the base on every process start. A profile is selected only when all four pointers resolve to the same base. If they do not, the exploit stops instead of continuing with an uncertain profile.

A different libvips version or build would require additional profiling. The relative positions of the leak anchors, the `memcpy` GOT entry, and the internal loader routine may move even though the overall exploitation technique remains the same. A new target therefore needs its own binary fingerprint and profile, with the remotely leaked values validated against that build before exploitation.

The following simplified excerpt shows the core calculation. It extracts eight-byte values from the returned alpha pixels, subtracts every declared anchor, and counts only aligned candidates that meet each anchor's repetition threshold:

```
alpha = image.getchannel("A").tobytes()
values = {
    int.from_bytes(alpha[offset:offset + 8], "little")
    for offset in range(0, len(alpha) - 7, 8)
}

for value in pointer_values:
    for anchor in profile.libvips.leak.anchors:
        candidate = value - anchor.offset
        alignment = profile.libvips.leak.base_alignment
        if candidate > 0 and not (candidate & (alignment - 1)):
            repetitions[candidate, anchor.offset] += 1

anchors_by_base = defaultdict(set)
for (base, anchor_offset), hits in repetitions.items():
    anchor = next(
        item for item in profile.libvips.leak.anchors
        if item.offset == anchor_offset
    )
    if hits >= anchor.minimum_repetitions:
        anchors_by_base[base].add(anchor_offset)

supported = [
    base for base, matched_anchors in anchors_by_base.items()
    if len(matched_anchors) >= profile.libvips.leak.required_anchors
]
```

The exploit proceeds only if the complete manifest produces exactly one supported profile and base pair. It then adds the selected profile's `memcpy@GOT` and internal `GModule-loader` offsets to that base. This is how the final chain handles a PIE Node executable: it never needs a Node address. Every runtime code address used by the overwrite belongs to `libvips` and is calculated from the recovered `libvips` base.

Which values are hardcoded?

The CVE-2026-32740 exploit needs several hardcoded profile values. The Linux distribution and `glibc` version alone are not enough. The values depend on the complete native stack. We keep them in a versioned native-stack profile instead of scattering constants through the exploit code. Each value comes from a specific part of the tested stack:

- The `libvips` leak profile declares the shared anchors `0x1be1b0`, `0x1c3120`, and `0x1c3130`. Its fourth anchor identifies the surrounding stack: `0x33cd0c` for Ubuntu and `0x1de9c0` for Debian.
- `libvips.memcpy_got.offset = 0xf83098` is the `memcpy` GOT offset in that same `libvips` binary. The runtime address is the leaked `libvips` base plus this offset.
- `libvips.control_target.offset = 0x3e995e` locates the internal `g_module_open_full` routine in the same `libvips` image. The runtime target is the leaked `libvips` base plus this offset. Because this internal routine is not exported as a dynamic symbol, the profile stores its first 32 bytes and the offline verifier checks those exact bytes.
- `heap_calibration.anchor_page_low16 = 0x6000` is the verified allocator page lane shared by the two tested PIE layouts.
- `heap_calibration.fake_node_delta_from_anchor_page` is `-0x690` for Ubuntu and `-0x3a0` for Debian. Those relations produce the `0x5970` and `0x5c60` partial-pointer selectors respectively.
- The profile's 116 by 33 tile geometry and 64-byte canvas stride create the required 58-byte row and place the forged structure at the measured location.

The randomized `libvips` base and runtime loader address are not hardcoded. The exploit recovers the base from each process's returned pixels, then calculates the loader as `libvips base + 0x3e995e`. The two-byte heap selector is a separate allocator-layout invariant in the exact PIE profile. A matching distribution or `glibc` version is not enough by itself; a different Node, sharp, `libvips`, `libheif`, compiler, or allocator layout can change one or more relative values and therefore needs its own verified profile.

How the exploit derives the heap selector

This is where the two PIE profiles differ. Both measured layouts place the relevant allocator page in low-16 lane `0x6000`, but the forged node sits at a different offset from that page. The exploit calculates the selector from the profile relationship:

```
page_lane          = 0x6000
ubuntu_selector    = (page_lane - 0x690) & 0xffff # 0x5970
debian_selector    = (page_lane - 0x3a0) & 0xffff # 0x5c60
```

The profile marker in the returned PNG chooses the native stack, and that profile supplies the matching page-to-node relation. The Debian relation was measured unchanged across five fresh diagnostic process lifetimes before it was added to the manifest. The selector is therefore a property of an exact verified native stack, not a value inferred from an OS name alone.

3. Turn the overflow into a GOT overwrite

How libheif stores image planes

`heif_channel` is an enum that names an image component. In this build, `heif_channel_Y` is 0, `heif_channel_Cb` is 1, and `heif_channel_Cr` is 2. libheif uses that enum as the key in `std::map<heif_channel, ImagePlane> m_planes`.

`ImagePlane` is libheif's metadata for one component's pixel buffer. It records the plane's dimensions, bit depth, allocation details, `mem`, and `stride`. The `mem` field points to the aligned start of the usable pixel buffer. The `stride` field is the number of bytes from the start of one row to the start of the next. Stride can be larger than the visible row width because an allocator may add alignment padding.

```
row_address    = mem + (row_number * stride)
sample_address = row_address + (column * bytes_per_sample)
```

`std::map` is an ordered key-value container. The C++ standard does not require one particular internal data structure, but the tested GNU libstdc++ implementation stores this map in `std::_Rb_tree`, a red-black tree. A red-black tree is a self-balancing binary search tree. Each node holds a key-value pair, a colour bit, and links to its parent, left child, and right child. The colour rules keep the tree approximately balanced, while a lookup compares the requested key and follows left or right links until it finds a match.

Here, the key is a `heif_channel` value and the value is its `ImagePlane`. When the article refers to the node's right-child pointer, it means the right link in this GNU tree node, not anything inside the Node.js runtime. Corrupting that link changes which key-value entry the next `get_plane(Cr)` search visits.

The 58-byte overwrite

The overflowing Cb allocation is next to the map's live root node in the measured heap layout. Those 58 controlled bytes have this job:

```
overflow +0x00..0x0f  allocation tail, kept zero
overflow +0x10..0x1f  next glibc chunk header, prev_size=0, size=0x75
overflow +0x20..0x37  first 24 bytes of the live tree node, kept zero
overflow +0x38..0x39  profile selector, 0x5970 on Ubuntu or 0x5c60 on Debian
```

To avoid an immediate allocator rejection, the payload rebuilds the glibc header with its expected `0x75` size field, including the `PREV_INUSE` bit. It also recreates the first tree fields.

Only the final two bytes change the root node's right-child pointer. Keeping the upper six pointer bytes untouched preserves the randomized heap prefix. Changing the low 16 bits selects a nearby address in the same 64 KiB heap region.

The forged plane-map node

That nearby address is not accidental. Earlier Cb rows have already copied attacker-controlled pixels there. The exploit arranges those bytes as a forged `std::map` node with the key `heif_channel_Cr`. Its embedded `ImagePlane` has a 58-byte width, a destination pointer at `memcpy@GOT - 16`, and a zero row stride:

```
struct.pack_into("<I", node, 32, 2)           # key: Cr
struct.pack_into("<II", node, 48, 58, 66)      # plane dimensions
struct.pack_into("<Q", node, 64, write_target) # ImagePlane::mem
struct.pack_into("<I", node, 88, 0)           # ImagePlane::stride
```

This is what the earlier wording “fake plane descriptor” meant. More precisely, it is a forged plane-map node, stored in valid image pixels, whose value looks like an `ImagePlane`. When libheif next calls `get_plane(Cr)`, the corrupted right-child pointer steers the tree search to the forged node. The matching Cr key makes the lookup return the forged `mem` and `stride` fields.

Why this is a write-what-where

The Cr copy loop now treats `memcpy@GOT - 16` as the output plane. Since the forged stride is zero, every output row starts at the same address. Consequently, the pixels are the “what” and `ImagePlane::mem` is the “where”, which is the constrained write-what-where primitive used by the exploit.

The Global Offset Table (GOT) holds the runtime destinations of imported functions. Calls through `memcpy@GOT` normally resolve to glibc's `memcpy`. Replacing that entry changes the destination of the next indirect call at the same copy site.

4. Load the uploaded native library

The first controlled row

Each controlled Cr row has a simple layout. Bytes 0 to 15 contain the null-terminated library path `uploads/x.jpg`. The next eight bytes contain the runtime address calculated as `libvips base + 0x3e995e`. Everything from byte 24 to byte 57 is zero. Since the destination begins 16 bytes before the GOT entry, the first copy leaves the path immediately before the slot and writes the GModule loader address into `memcpy@GOT`.

Why the helper fits the call site

The selected routine is libvips' internal copy of `g_module_open_full`. Conceptually, it receives a library path, loader flags, and an optional `GError **`. It checks the named file and calls `dlopen`, which runs the shared object's constructor as soon as the load succeeds.

On x86-64, the hijacked `memcpy(dest, src, len)` call has `dest` in `RDI`, `src` in `RSI`, and `len` in `RDX`. After the first controlled copy, `RDI` points to `memcpy@GOT - 16`, which now contains `uploads/x.jpg`. That is exactly where the `GModule` routine expects its path.

The other two registers do not contain normal `GModule` arguments. `RSI` still holds an image-row pointer and `RDX` still contains the 58-byte copy length. This exact loader implementation masks `ESI` down to the supported `GModule` flag bits. On a successful load, it never dereferences `RDX` as an error pointer. We verified that exact success path against the tested `libvips` binary before using it in the remote chain.

The first `memcpy` finishes after replacing its own `GOT` entry. The loop then processes the next `Cr` row through the same indirect call site. This time execution enters the `GModule` loader with `RDI` pointing to the stored path. The loader opens `uploads/x.jpg`, its constructor runs `/usr/bin/id`, and the result is sent to the listener.

Loader entry: `RIP` is `libvips` plus `0x3e995e`, and `RDI` points to `uploads/x.jpg`.

`dlopen` receives `uploads/x.jpg`; frame 1 returns into `libvips`' `GModule` loader.

Measured results

We tested the complete exploit against 10 newly started Ubuntu Node processes and 10 newly started Debian Node processes. All 20 runs reached command execution and returned the output of `/usr/bin/id`. `ASLR` selected a different `libvips` address in every process, and the exploit calculated the corresponding `GModule`-loader address each time.

Before creating the final payload, the exploit may need several `ASLR`-leak attempts. Each attempt uploads the leak `AVIF`, processes it through the public optimisation route, and checks the returned `PNG` for four pointers that identify one supported profile and one `libvips` base. If the evidence is incomplete or ambiguous, the exploit retries instead of guessing. Ubuntu needed between 3 and 23 attempts, while Debian needed between 3 and 15.

The observed result was 20 successful runs out of 20 in the two tested environments. This does not mean the exploit will work against every sharp deployment. Different versions of Node, sharp, `libvips`, `libheif`, `glibc`, or `libstdc++`, and changes to the allocator or upload path, may require a new profile.

The project also has 96 automated regression tests. These are code-level checks, not 96 additional exploit runs. They test profile validation, returned-pointer classification, heap calibration, address calculation, payload construction, and safe failure when the evidence does not match a supported target.

Versioned native-stack profiles

The build-specific values live in strict manifests. `profiles/native_stack_profiles.json` describes the original `ET_EXEC` lab. `profiles/native_stack_profiles_pie.json` contains the Ubuntu and Debian `ET_DYN` profiles. Each profile records the exact Node, Next.js, sharp,

libvips, libheif, glibc, and libstdc++ identities, plus the loader, relocation, returned-pointer anchors, C++ layouts, tile geometry, and allocator relationship.

A separate offline verifier checks the supplied binaries and package manifests before that profile is used. It validates SHA-256 hashes, ELF build IDs and type, the hidden loader offset and bytes, the `memcpy` jump slot, npm package versions, and the bundled libvips and libheif versions. The runtime profile loader also rejects unknown fields, missing values, unsafe paths, and inconsistent geometry before opening the callback listener or sending an HTTP request.

The exploit waits until four returned pointers identify exactly one supported profile and one libvips base. The selected profile then supplies the matching page-to-node relationship. If the evidence is incomplete or ambiguous, the exploit retries the ASLR-leak request before creating the final payload. The repository contains the exploit, the exact profiles, the automated tests, and the results from each of the 20 end-to-end runs.

Reproducing the proof in the authorised lab

The private repository contains a minimal reproducible lab, the native-stack profile and verifier, and one end-to-end exploit script. After installing the documented prerequisites, start the target:

```
cd lab
npm ci
npm run build
npm run start
```

Then run the exploit from the repository root:

```
python3 exploit.py \
  --target http://127.0.0.1:3000 \
  --callback-host 127.0.0.1 \
  --profile-manifest profiles/native_stack_profiles_pie.json \
  --json-output result.json \
  --concise
```

The PIE exploit selected the Ubuntu profile, recovered the libvips base, and returned the target's `/usr/bin/id` output.

Those 20 end-to-end acceptance runs each selected the correct profile, identified a different libvips base, calculated `libvips + 0x3e995e`, derived the profile's heap selector, uploaded `x.jpg`, generated the runtime AVIF, and returned the target's real `/usr/bin/id` output.

Deployment conditions and limitations

This proof demonstrates impact, not universal exploitability. A target must expose both a compatible vulnerable image-processing path and a way to preserve an attacker-controlled shared object, submitted as an image, at a short path reachable by the native loader.

The libvips loader offset, relocation offset, allocator page lane, and partial-pointer relationship are specific to a tested native stack. The loader target does not depend on the Node executable address, but a different Node or allocator layout can change the relation used by the forged-node selector. Such a deployment needs its own verified profile.

Mitigating CVE-2026-32740

- Upgrade to libheif 1.22.0 or later, then rebuild or replace every sharp and libvips package that bundles the vulnerable library.
- Confirm the native library actually loaded in production. Updating only a top-level JavaScript package may not replace a bundled binary.
- Disable or isolate AVIF and HEIF processing until the deployed native dependency chain has been verified.
- Validate file magic, generate server-side filenames, and keep uploads outside the application working directory on a `noexec` filesystem where possible.
- Run image decoders in disposable, least-privileged workers without application secrets or unrestricted outbound network access.
- Alert on native image-worker crashes, unexpected module loads, or executable content opened from upload directories.

Conclusion

For us, the interesting part is how these primitives fit together. CVE-2026-32740 supplies the adjacent write, a separate returned-pixel disclosure defeats ASLR, the forged plane-map node turns the corruption into a constrained write-what-where, and the immediate `memcpy@GOT` transition reaches a libvips-relative native loader before another worker can intervene.

Across the two tested PIE labs, this chain produced 20 callbacks in 20 fresh processes, with 20 distinct randomized libvips bases. Every callback returned the output of `/usr/bin/id`. This is more than a decoder crash. Under the documented application conditions, an unauthenticated image upload reaches native command execution even though the Node executable is position independent.

Our PoC is available in the [Fortbridge libheif-grid Next.js RCE repository](#).

References

- [libheif security advisory for CVE-2026-32740](#)
- [Hacktron, Hacking OpenAI through its Discourse deployment](#)
- [sharp project repository](#)

- [Fortbridge private exploit repository](#)
- [GLib GModule documentation](#)
- [GNU libstdc++ `std::map` implementation](#)

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